

## 5.1.3 Data Analytics & Components

ڈیٹا اینالیٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلو...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 04 of 12

### 5.1.3 Data Analytics & Components

#### LEARN IT

**Data Analytics** is the process of **collecting, organising, cleaning, studying and presenting data** to find useful information. It helps people make better decisions.

**Data** means raw facts and figures (for example, "72, 45, 88"). **Information** is data that has been processed so it has meaning ("Ali scored 72%; he passed"). Data Analytics is the bridge from data to information.

**Source example:** A college collects exam marks, calculates percentages, and identifies which students need extra help. This entire process is Data Analytics.

#### Components of Data Analytics

Data Analytics comprises **five key components**, which happen in order like a pipeline:

1. Collection → 2. Storage → 3. Cleaning → 4. Analysis → 5. Visualization

1. **Data Collection** — Gathering raw information from multiple sources: websites, surveys, sensors, hospital records, bank transactions.
2. **Data Storage** — Storing data safely and properly after collection. It can be stored in files, databases, spreadsheets or the cloud.
3. **Data Cleaning** — Identifying and correcting errors in data. Raw data may contain:
  - **Duplication** of records (the same student entered twice),
  - **Incorrect values** (typos, e.g. marks "870" instead of "87"),
  - **Missing** files or records (a student's marks not entered),
  - **Improper formats** (dates written as 05/10/2026 in one place and "5 Oct 26" in another).
4. **Data Analysis** — Studying the clean data to identify patterns, trends and relevant information for problem-solving (e.g. finding out which subjects students failed the most).
5. **Data Visualization** — Creating visual representations of data (plots, charts, graphs) to provide valuable insights at a glance (e.g. a pie chart showing pass vs fail percentage).

### WORKED EXAMPLE

A college in Sukkur analyses Physics test results (out of 50) for 6 students.

**Step 1 — Collection:** Marks are gathered from the teacher's register.

| Roll No | Name   | Marks (raw) |
|---------|--------|-------------|
| 1       | Ayesha | 42          |
| 2       | Bilal  | 18          |
| 3       | Hina   | 35          |
| 3       | Hina   | 35          |
| 4       | Kashif | 250         |
| 5       | Sana   | (missing)   |

**Step 2 — Storage:** The marks are typed into a spreadsheet saved on Google Drive (cloud).

**Step 3 — Cleaning:** Hina's record is a *duplicate* → delete one copy. Kashif's 250 is an *incorrect value* (more than 50) → the register shows 25, so correct it. Sana's marks are *missing* → ask the teacher; she was absent, so mark her "absent".

**Step 4 — Analysis:** Clean marks: 42, 18, 35, 25. Pass mark is 17 (33% of 50 = 16.5, rounded up). All 4 passed. Percentages: Ayesha 84%, Bilal 36%, Hina 70%, Kashif 50%. Average =  $(42 + 18 + 35 + 25) \div 4 = 120 \div 4 = 30$  marks = 60%. Bilal is closest to failing → needs extra help.

**Step 5 — Visualization:** A bar chart shows each student's percentage; Bilal's short bar makes the problem visible at a glance.

### REMEMBER IT

**"Cows Sit Calmly And Vote"** — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar (store), pick out stones (clean), cook (analyse), serve beautifully on a plate (visualise).

### COMMON MISTAKE

**Mistake:** Skipping cleaning and going straight to analysis. **Fix:** Dirty data gives wrong answers. In the example, Kashif's wrong "250" would have made the average  $(42 + 18 + 35 + 250) \div 4 = 86.25$  — more than 50, which is impossible. Always clean first.

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔ اس کے پانچ اجزاء ہیں: ڈیٹا جمع کرنا، محفوظ کرنا، صاف کرنا (غلطیاں اور دہرائے گئے ریکارڈ ہٹانا)، تجزیہ کرنا، اور چارٹ یا گراف کی صورت میں دکھانا۔

#### CHECK YOURSELF

1. List the five components of Data Analytics in order.
2. Name four kinds of errors removed during data cleaning.
3. A pie chart of pass/fail percentage belongs to which component?

Answers: 1. Collection, Storage, Cleaning, Analysis, Visualization. 2. Duplicate records, incorrect values (typos), missing records, improper formats. 3. Data Visualization.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Data analytics is:

- a. Collect, clean, study, present data for decisions
- b. Only drawing NAND
- c. Only cabling
- d. Only Figma

**MCQ 2.** Cleaning data means:

- a. Fixing missing/wrong values
- b. Washing the PC
- c. Defrag only
- d. Sorting OSI layers

**MCQ 3.** A dashboard is:

- a. A visual summary of analytics
- b. A primary key
- c. A sprint log only
- d. A K-map

**MCQ 4.** Analytics without a question is:

- a. Just a pile of charts
- b. Always AI
- c. Always IoT
- d. Always SQL keys

**MCQ 5.** Finding that "second shift fails more in CS" is:

- a. Insight from analysis
- b. A foreign key
- c. A gate delay
- d. A beachhead

**MCQ 6.** Python's later tool for this in XII is:

- a. Pandas + charts
- b. LogiSim
- c. Figma only
- d. Access macros only

## Extra short questions

1. Define data analytics and list its steps.
2. Why clean first?
3. Give a college decision driven by analytics.
4. How is analytics different from IoT?

## Extra long question

1. A principal has 3 years of BIEK results. Outline an analytics project: source, clean, one chart, one decision.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

### **Short**

1. Collect → organise/clean → analyse → present → decide.
2. Garbage in, garbage out — mean of marks with blanks is wrong.
3. Extra classes for a subject whose fail rate jumped.
4. IoT gathers live device data; analytics is the studying/presenting (often of that data).

### **Long**

1. Source: gazette CSVs. Clean: missing roll numbers. Chart: line of pass% by year.  
Decision: change lab hours if practical marks drag theory. Mention ethics (no public shaming of named students).

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Next lecture: Table: IoT vs AI vs Data Analytics Comparison